

Individual Claim Reserving

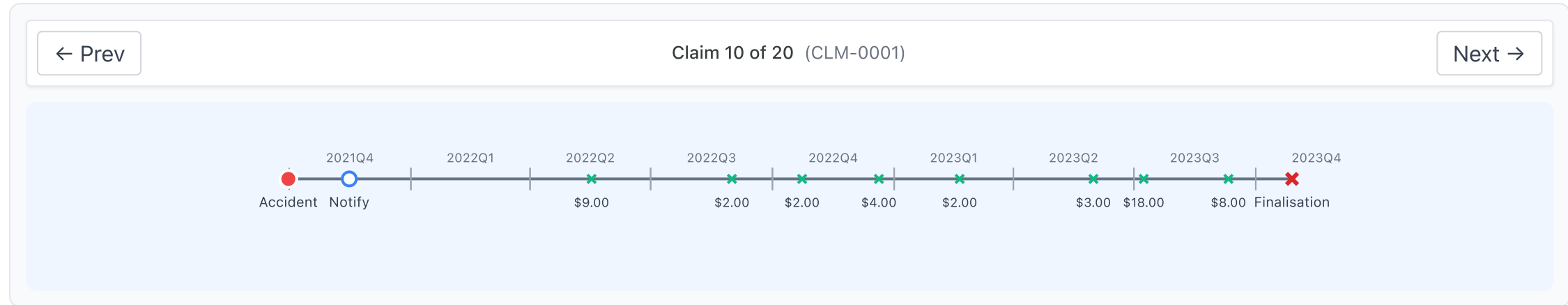
ACTL3143 & ACTL5111 Deep Learning for Actuaries
Patrick Laub

Lecture Outline

- **Introduction**
- Insurance Claim Payments and Case Estimates
- Benchmark: Aggregate Reserving
- Individual Claim Modelling Choices
- Preprocessing One Claim
- Splitting Claims by Finalisation Quarter
- Splitting Claims by Notification Quarter
- Case Study: Synthetic Claims
- Fitting the Models
- Metrics
- Did the Model Beat the Assessors?
- Wrapping Up

Individual claim reserving

An accident has occurred, the insurer has been notified, and maybe parts of the claim have already been paid, now we need to guess how much remains in this realised claim.



This is potentially a promising application for neural networks.

Ultimate claim size

For claim k , we want to know the *ultimate claim size*

$$U_k := \sum_t P_{k,t},$$

This is the sum of all *incremental payments* $P_{k,t}$ (also called *partial payments*) over the complete lifetime of the claim.

i Assumption: payments are positive

We assume $P_{k,t} \geq 0$ throughout. In reality some payments are *negative* — called *recoveries* (e.g. money coming back from a third party or a salvaged vehicle). They are typically so rare and small relative to normal payments that ignoring them is reasonable (though this depends on the dataset).

Outstanding claim amount

Say the valuation date is v , then we've seen the history of the claim $\mathcal{H}_{k,v}$ to that time, including past payments.

So the *outstanding* (future unpaid amount) for claim k is

$$O_k(v) := \sum_{t > v} P_{k,t}.$$

Writing $P_k(v) := \sum_{t \leq v} P_{k,t}$ for the *cumulative paid to date*, if the claim is still ongoing then

$$U_k = P_k(v) + O_k(v).$$

In words:

Ultimate = Cumulative payments before v + Outstanding amount from v onwards

Total outstanding

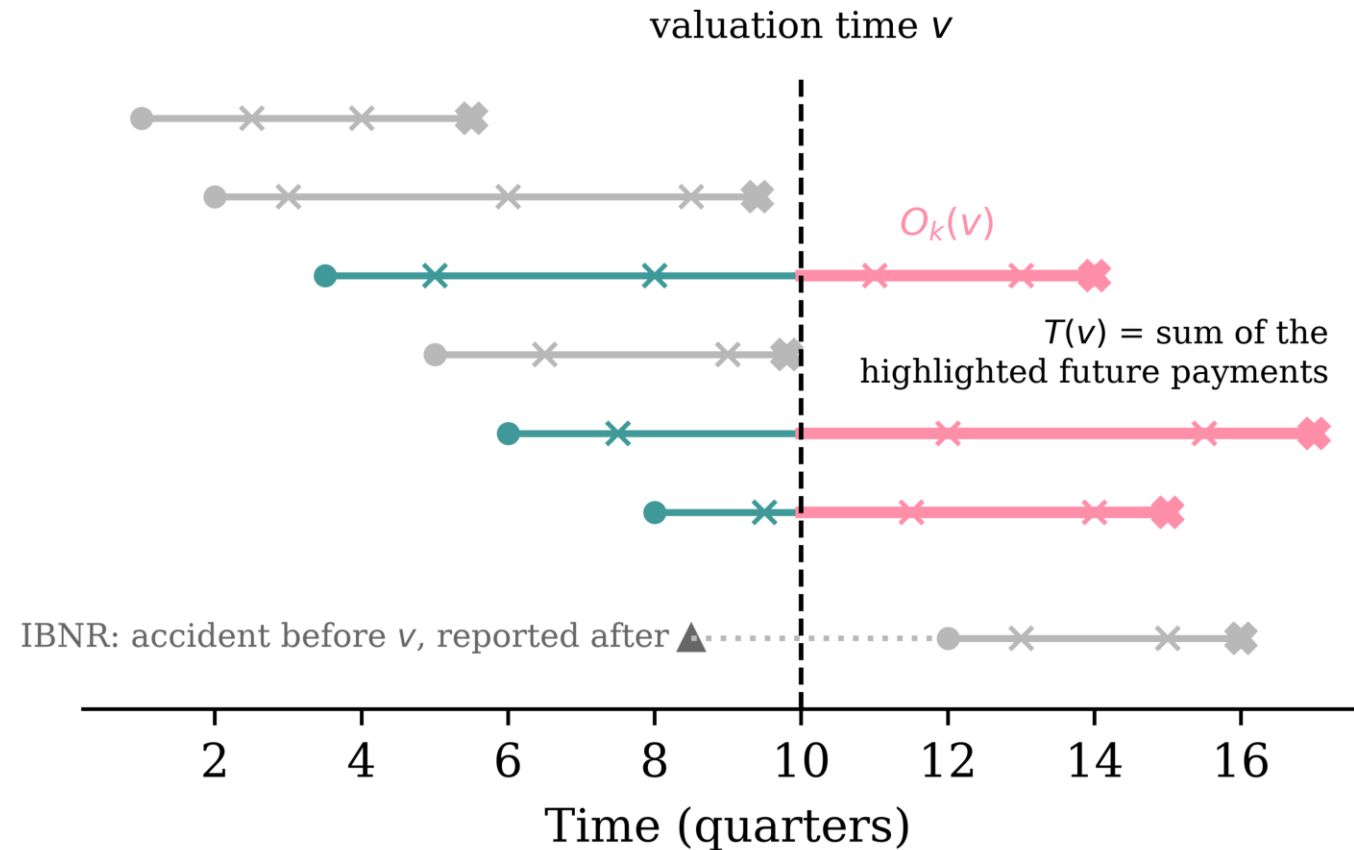
For a portfolio, we'd like to know the *total outstanding*

$$T(v) := \sum_{k \in \mathcal{O}(v)} O_k(v) ,$$

where $\mathcal{O}(v)$ is the set of all open claims at valuation time v .

Note: this doesn't include *incurred but not reported (IBNR)* claims.

The total outstanding, visually



Each open claim's future payments (highlighted) get summed to give $T(v)$.

The bottom claim has *occurred* before v but hasn't been *notified* yet. No individual model can reserve for a claim it cannot see — IBNR needs a separate adjustment.

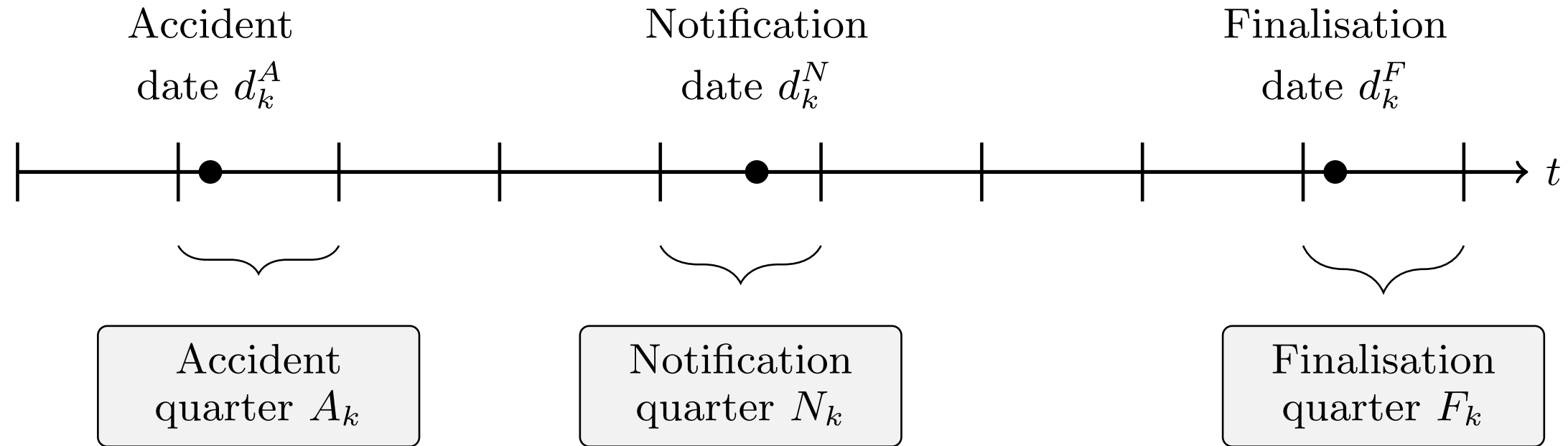
Why is pricing different from reserving?

- *Pricing* is prospective and per-policy: no claim exists yet, and we predict the frequency/severity of hypothetical future claims from policy covariates (recall the French motor lectures).
- *Reserving* is about claims that have already happened: we condition on each claim's unfolding history — payments to date, case estimates — to predict what is still to be paid.
- Consequence for the ML setup: pricing rows are policies with static features; reserving rows are claim-quarter snapshots with *time-varying* features. That's why this lecture needs a whole preprocessing pipeline.

Lecture Outline

- Introduction
- **Insurance Claim Payments and Case Estimates**
- Benchmark: Aggregate Reserving
- Individual Claim Modelling Choices
- Preprocessing One Claim
- Splitting Claims by Finalisation Quarter
- Splitting Claims by Notification Quarter
- Case Study: Synthetic Claims
- Fitting the Models
- Metrics
- Did the Model Beat the Assessors?
- Wrapping Up

The key dates



Key dates

A claim is *open* if not yet finalised.

$$\mathcal{O}(v) := \{k : d_k^F > v\}.$$

(Though, technically, claims sometimes finalise, and then get reopened with further payments, and refinalised again.)

Series of incremental payments

Event	Date	Amount
Accident	2021-09-30	
Notification	2021-11-15	
Payment #1	2022-05-17	\$9.00
Payment #2	2022-08-31	\$2.00
Payment #3	2022-10-23	\$2.00
Payment #4	2022-12-20	\$4.00
Payment #5	2023-02-19	\$2.00
Payment #6	2023-05-31	\$3.00
Payment #7	2023-07-08	\$18.00
Payment #8	2023-09-10	\$8.00
Finalisation	2023-10-28	

Case estimates

Alongside payments, the insurer holds a *case estimate* $C_{k,v} \geq 0$: a claims assessor's live, subjective estimate of the outstanding liability at the end of quarter v .

The assessor is aiming for

$$C_{k,v} \approx O_k(v) .$$

The crucial difference: $C_{k,v}$ is *known at time v* , whereas $O_k(v)$ is only knowable in retrospect, once the claim finalises.

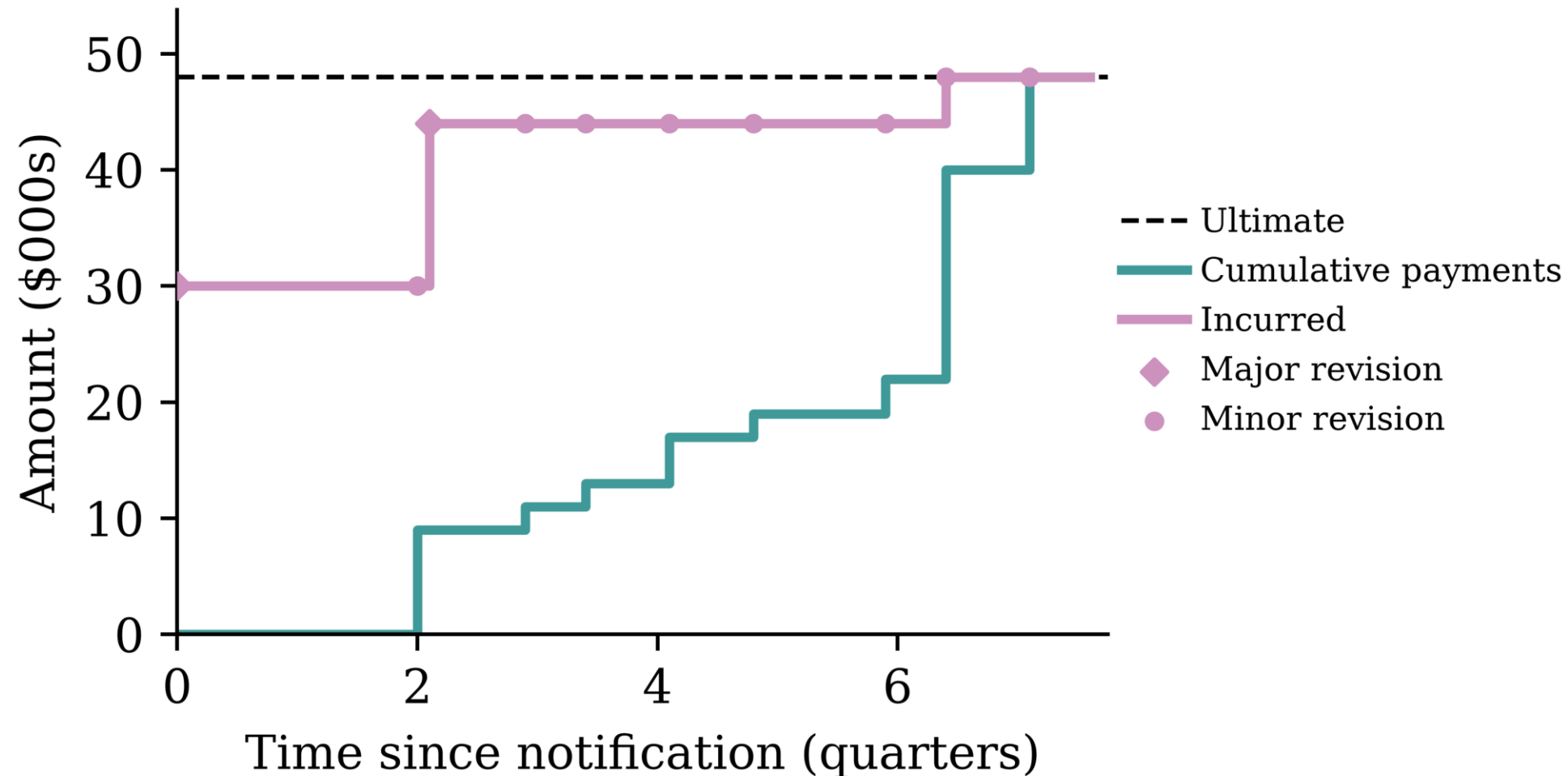
So case estimates play two roles for us:

- a *strong predictor* to feed into the model, and
- a *benchmark to beat* (if the model can't out-predict the assessors, why bother?).

Payments and case estimates together

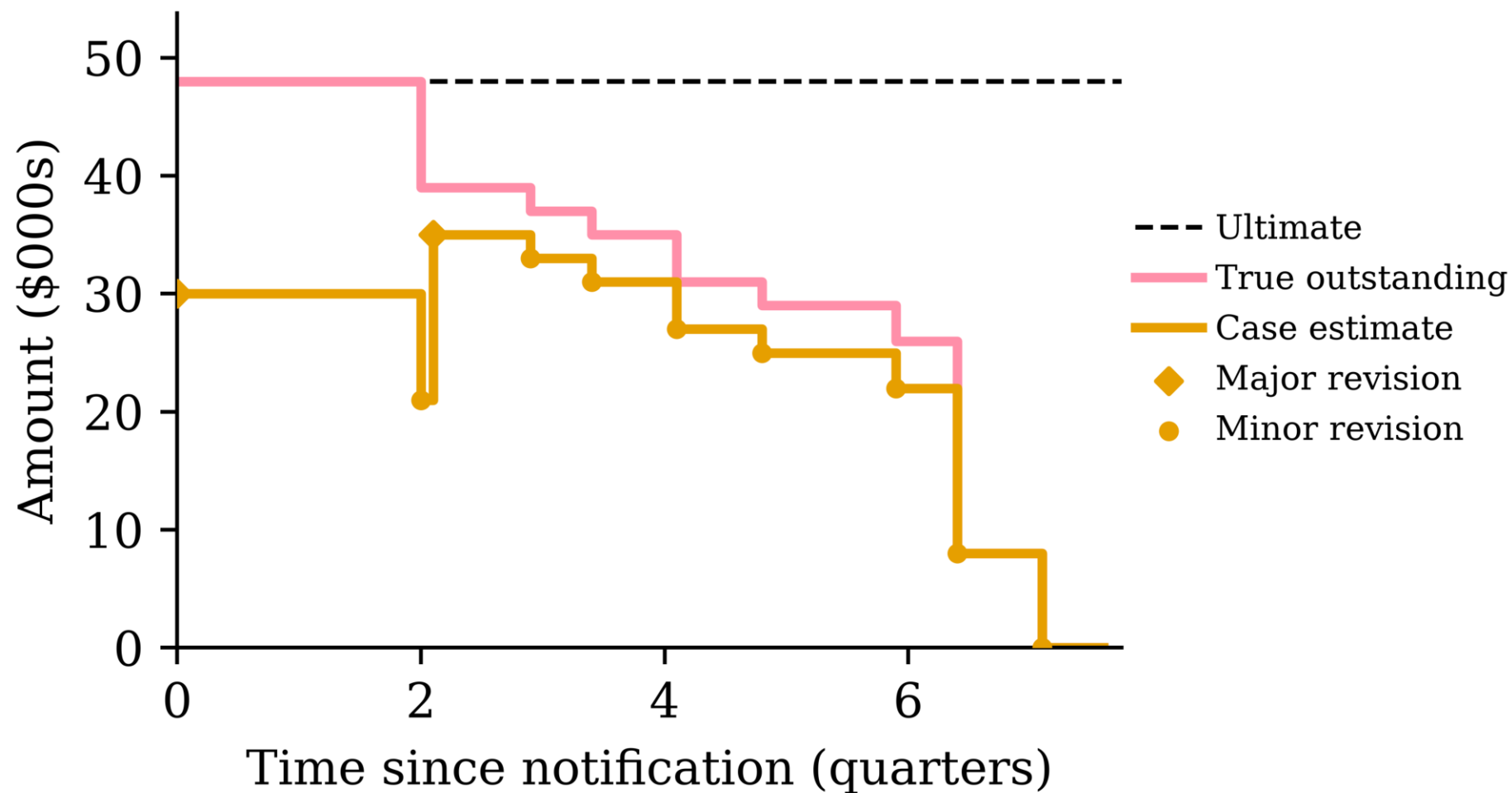
Date	Event	Payment	Case estimate
2021-11-15	Notification, initial estimate		\$30.00
2022-05-17	Payment #1	\$9.00	\$21.00
2022-06-20	Review: worse than hoped		\$35.00
2022-08-31	Payment #2	\$2.00	\$33.00
2022-10-23	Payment #3	\$2.00	\$31.00
2022-12-20	Payment #4	\$4.00	\$27.00
2023-02-19	Payment #5	\$2.00	\$25.00
2023-05-31	Payment #6	\$3.00	\$22.00
2023-07-08	Payment #7 (settlement agreed)	\$18.00	\$8.00
2023-09-10	Payment #8	\$8.00	\$0.00
2023-10-28	Finalisation		\$0.00

From the policyholder's perspective



Payments step up toward the ultimate. The *incurred*, which is cumulative payments plus the case estimate of the outstanding, is the assessor's running view of the ultimate.

From the insurer's perspective



The true outstanding $O_k(v)$ — the *outstanding claims liability (OCL)* — and the *case estimate* $C_{k,v}$ (the assessor's live forecast of it) both drop to zero.

Lecture Outline

- Introduction
- Insurance Claim Payments and Case Estimates
- **Benchmark: Aggregate Reserving**
- Individual Claim Modelling Choices
- Preprocessing One Claim
- Splitting Claims by Finalisation Quarter
- Splitting Claims by Notification Quarter
- Case Study: Synthetic Claims
- Fitting the Models
- Metrics
- Did the Model Beat the Assessors?
- Wrapping Up

Three levels of claim data granularity

1. *Individual transaction level*: Stores every transaction or other update for every claim. Highest granularity. Often just visible in the insured's bank statements, or in insurance company's finance records. Usually considered too granular to be useful for actuaries.
2. *Individual quarterly summaries*: This aggregates all the payments within a quarter (alternatively per month) for each claim. This is what the actuary would see.
3. *Portfolio quarterly summaries*: Aggregates all payments within a quarter (alternatively per month) across all claims. This feeds into a Chain Ladder style of reserving.

Aggregate reserving

You have seen the classical approach to reserving already. The insurer's past payments are aggregated into a *run-off triangle*: rows are accident years, columns are development years, and each cell holds the cumulative amount paid so far.

Accident Year	Dev 1	Dev 2	Dev 3
2021	100	150	165
2022	120	180	?
2023	140	?	?

The lower-right of the triangle is the future: it hasn't happened yet, and the *reserve* is our estimate of it.

Chain ladder

Chain ladder fills in the triangle by assuming each accident year develops in the same proportions as the past ones.

- Development factor from Dev 1 to Dev 2: $f_1 = \frac{150+180}{100+120} = 1.5$
- Development factor from Dev 2 to Dev 3: $f_2 = \frac{165}{150} = 1.1$

Projecting forward:

Accident Year	Dev 1	Dev 2	Dev 3	Reserve
2021	100	150	165	0
2022	120	180	<i>198</i>	18
2023	140	<i>210</i>	<i>231</i>	91

The total reserve here is $18 + 91 = 109$.

What chain ladder ignores

Every claim in the same accident year is treated identically. The triangle has already thrown away:

- *who* was injured (age, injury severity),
- *how* the claim is progressing (legal representation, payment pattern),
- the case estimates set by claims assessors.

If two claims have paid out the same amount so far, chain ladder implicitly reserves the same for both, even if one is a minor injury about to finalise and the other is heading to court.

Why individual claim reserving?

With claim-level data and machine learning, we can use those discarded covariates to predict each claim's outstanding amount separately.

- The portfolio reserve is then just a sum: $T(v) = \sum_{k \in \mathcal{O}(v)} O_k(v)$.
- Per-claim predictions are useful beyond the total: claims triage (which claims need senior assessors?), segmenting the reserve by injury severity or legal representation, spotting claims whose case estimates look off.
- Chain ladder remains the *benchmark*: an individual model that can't beat the triangle isn't worth its complexity.

Note

Next week's guest lecture digs into *when and why* aggregate models break down, and the trade-offs between micro- and macro-level models. Today we focus on individual-level approach and how to implement it.

Lecture Outline

- Introduction
- Insurance Claim Payments and Case Estimates
- Benchmark: Aggregate Reserving
- **Individual Claim Modelling Choices**
- Preprocessing One Claim
- Splitting Claims by Finalisation Quarter
- Splitting Claims by Notification Quarter
- Case Study: Synthetic Claims
- Fitting the Models
- Metrics
- Did the Model Beat the Assessors?
- Wrapping Up

A few major choices upfront

1. What are the model inputs and outputs?

Predictions are made from the perspective of some fixed time v , the *valuation quarter*.

The individual claim model has:

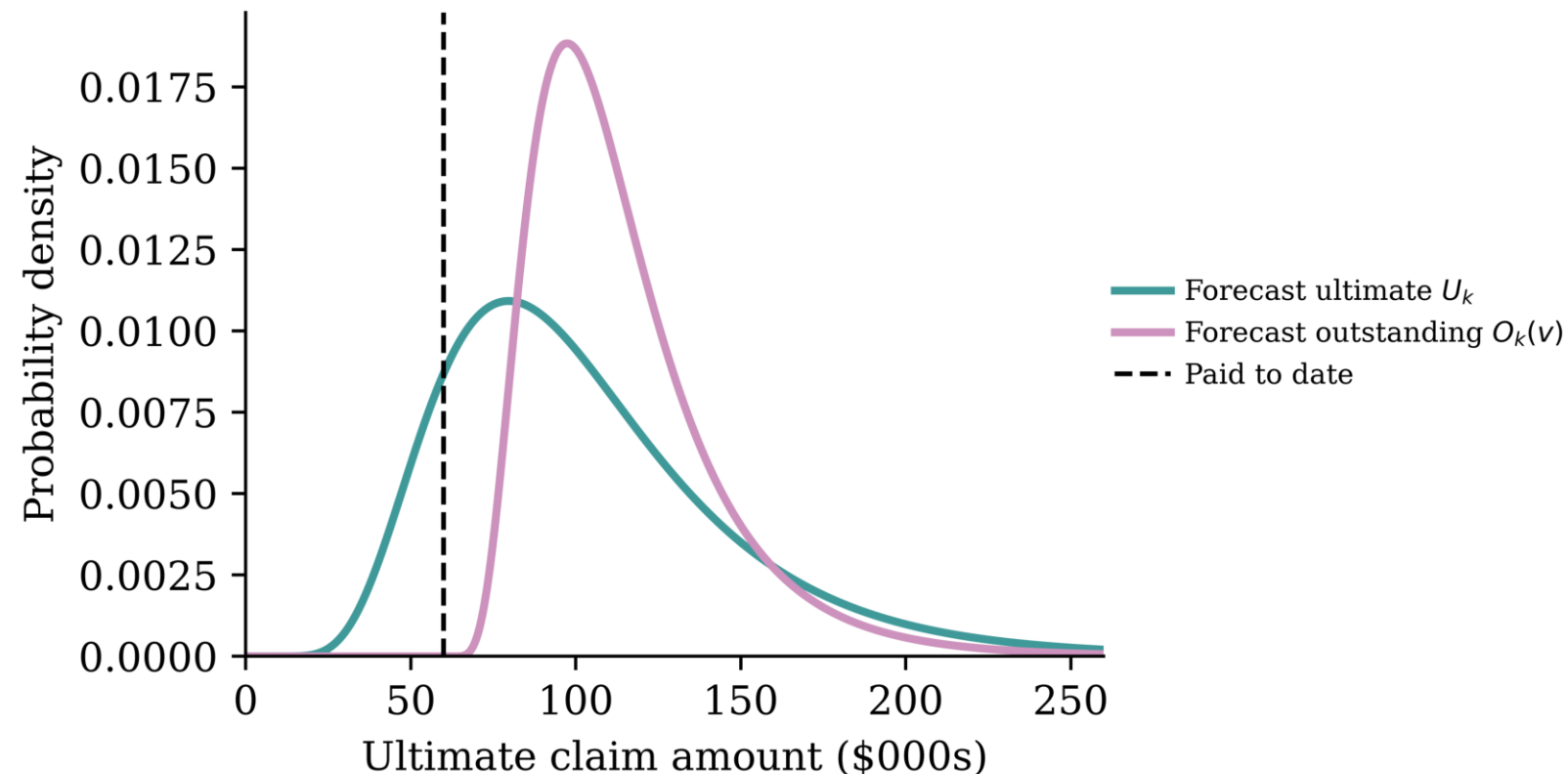
- *Input*: an $X_{k,v}$ summarising the claim's history up to v .
- *Output*: either the ultimate claim size U_k or the outstanding value $O_k(v)$.

2. Are we performing traditional regression or distributional regression?

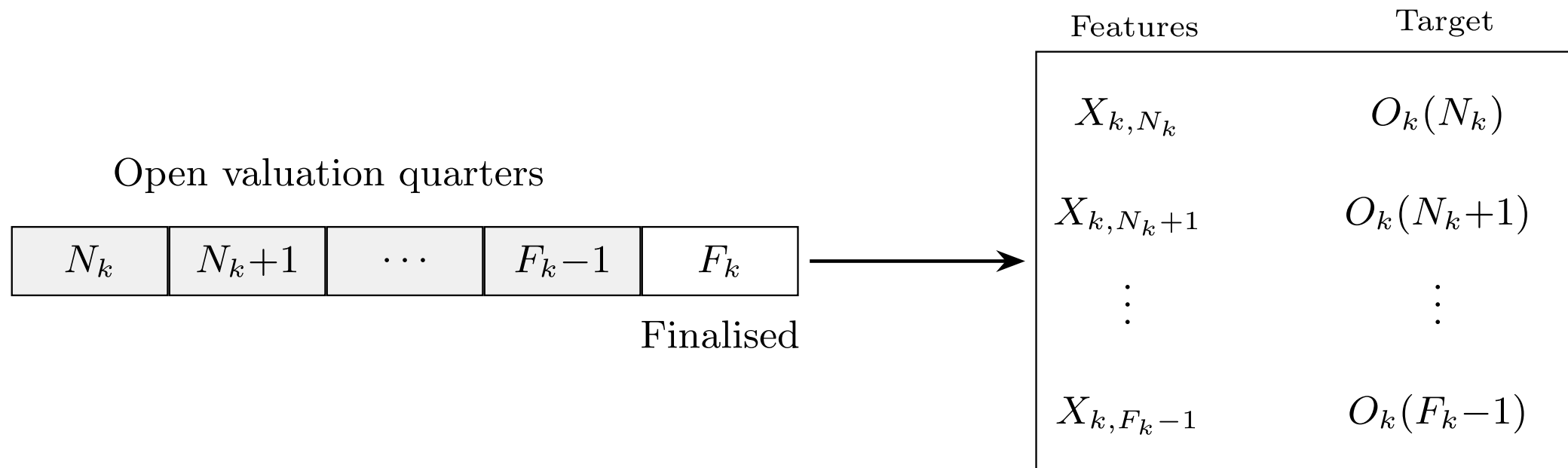
Two potential targets: ultimate or outstanding

The output distribution can sit on either of two quantities:

- the *ultimate* U_k , or
- the *outstanding* $O_k(v) = U_k - P_k(v)$.



Claim snapshots



We will turn one claim into multiple “claim snapshots”/rows of input.

One finalised claim, viewed retrospectively from *every* quarter it was open, yields one training example:

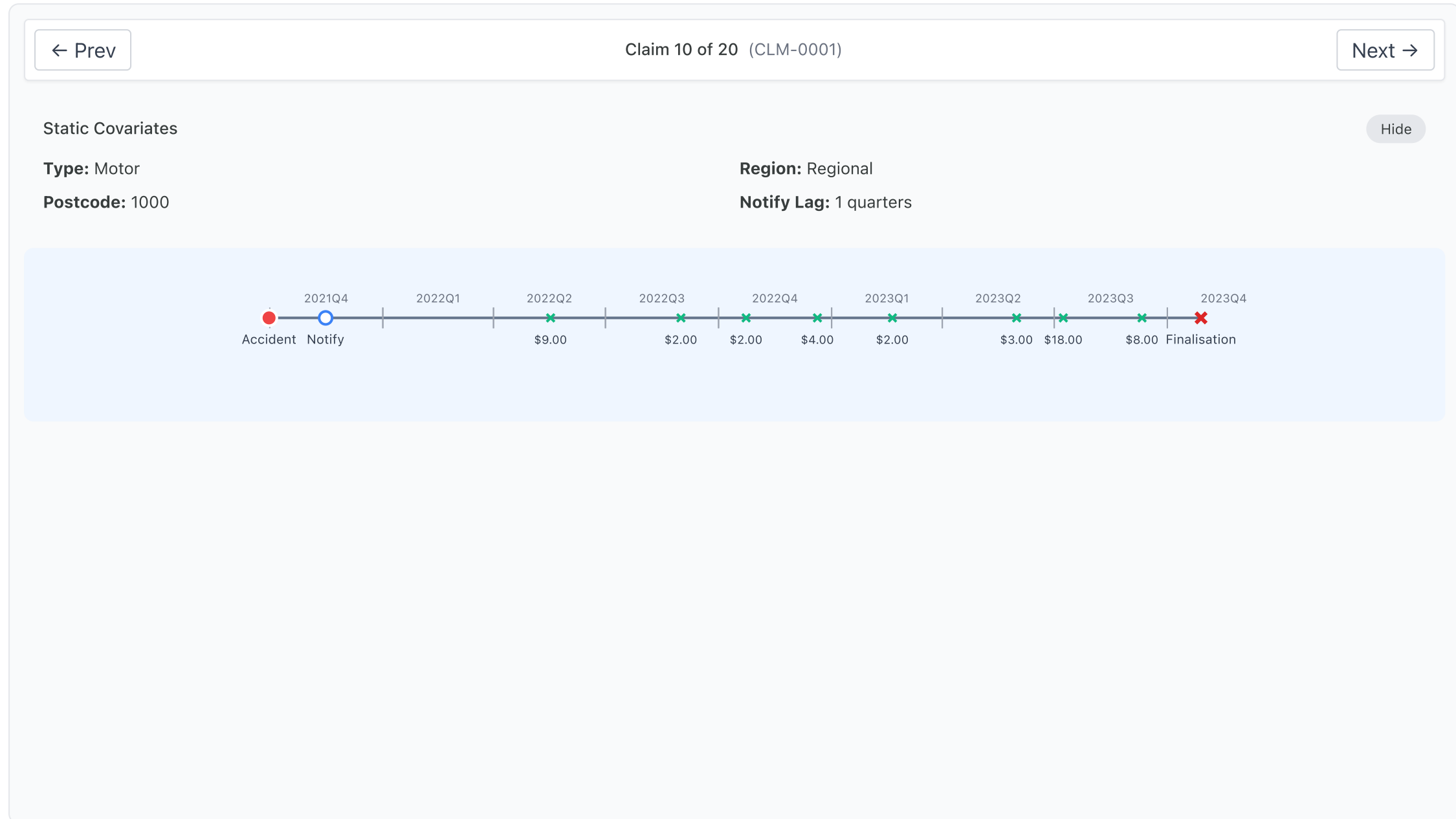
$$(X_{k,v}, O_k(v)) \text{ for } v = N_k, \dots, F_k - 1.$$

Note, claims with $N_k = F_k$ just disappear.

Lecture Outline

- Introduction
- Insurance Claim Payments and Case Estimates
- Benchmark: Aggregate Reserving
- Individual Claim Modelling Choices
- **Preprocessing One Claim**
- Splitting Claims by Finalisation Quarter
- Splitting Claims by Notification Quarter
- Case Study: Synthetic Claims
- Fitting the Models
- Metrics
- Did the Model Beat the Assessors?
- Wrapping Up

One claim's full history



Noting the quarters

Each event date is binned into its calendar quarter (A_k, N_k, F_k), and we also track the *development quarter* (quarters since the accident).

[< Prev](#)

Claim 10 of 20 (CLM-0001)

[Next >](#)

Event	Date	Calendar Quarter	Amount	Dev Quarter
Accident	2021-09-30	2021Q3		Q1
Notification	2021-11-15	2021Q4		Q2
Payment #1	2022-05-17	2022Q2	\$9.00	Q4
Payment #2	2022-08-31	2022Q3	\$2.00	Q5
Payment #3	2022-10-23	2022Q4	\$2.00	Q6
Payment #4	2022-12-20	2022Q4	\$4.00	Q6
Payment #5	2023-02-19	2023Q1	\$2.00	Q7
Payment #6	2023-05-31	2023Q2	\$3.00	Q8
Payment #7	2023-07-08	2023Q3	\$18.00	Q9
Payment #8	2023-09-10	2023Q3	\$8.00	Q9
Finalisation	2023-10-28	2023Q4		Q10

Quarterly aggregation

All payments within a quarter collapse into one number, $P_{k,t}$. Case estimates within a quarter are instead summarised by their *latest* value.

← Prev

Claim 10 of 20 (CLM-0001)

Next →

Payment Composition

Dev Q1	2021Q3	
Dev Q2	2021Q4	
Dev Q3	2022Q1	
Dev Q4	2022Q2	\$9.00
Dev Q5	2022Q3	\$2.00
Dev Q6	2022Q4	\$2.00 \$4.00
Dev Q7	2023Q1	\$2.00
Dev Q8	2023Q2	\$3.00
Dev Q9	2023Q3	\$18.00 \$8.00
Dev Q10	2023Q4	

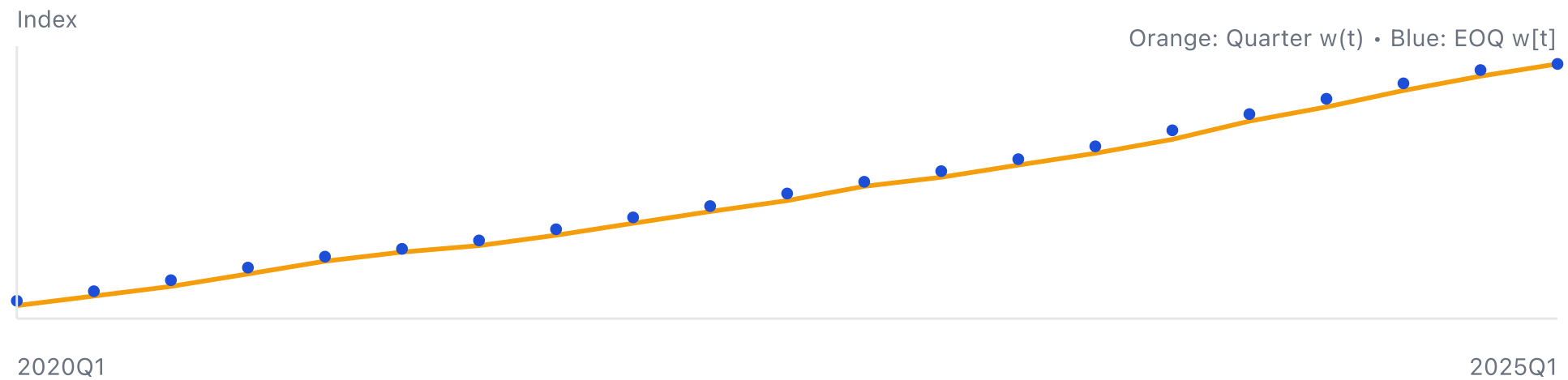
Quarterly Summary

Quarter	Sum
Dev Q1	\$0.00
Dev Q2	\$0.00
Dev Q3	\$0.00
Dev Q4	\$9.00
Dev Q5	\$2.00
Dev Q6	\$6.00
Dev Q7	\$2.00
Dev Q8	\$3.00
Dev Q9	\$26.00
Dev Q10	\$0.00

Adjusting for inflation

We need to adjust all monetary units to a common date to remove inflation.

WPI: Quarter vs End-of-Quarter



Adjusting for inflation, applied

Each of the selected claim's quarters gets an adjustment factor from the index.

[< Prev](#)

Claim 10 of 20 (CLM-0001)

[Next >](#)

Quarter & EOQ values and adjustment factors

Source Quarter	w(t) source	w[t] source	Target Quarter	w[T] target	Payments: w[T]/w(t)	Case est.: w[T]/w[t]
2021Q3	111.09	111.90	2025Q1	139.63	1.2569	1.2478
2021Q4	112.72	113.66	2025Q1	139.63	1.2387	1.2285
2022Q1	114.60	115.52	2025Q1	139.63	1.2184	1.2087
2022Q2	116.45	117.30	2025Q1	139.63	1.1991	1.1904
2022Q3	118.16	119.28	2025Q1	139.63	1.1817	1.1706
2022Q4	120.40	121.12	2025Q1	139.63	1.1597	1.1529
2023Q1	121.83	122.79	2025Q1	139.63	1.1461	1.1372
2023Q2	123.75	124.67	2025Q1	139.63	1.1283	1.1200
2023Q3	125.60	126.68	2025Q1	139.63	1.1117	1.1022
2023Q4	127.78	129.20	2025Q1	139.63	1.0927	1.0807

Quarter-Level Adjustments

Dev Quarter	Calendar Quarter	Nominal Sum	Adj Factor	Adjusted Sum
Dev Q1	2021Q3	\$0.00	—	\$0.00
Dev Q2	2021Q4	\$0.00	—	\$0.00
Dev Q3	2022Q1	\$0.00	—	\$0.00
Dev Q4	2022Q2	\$9.00	1.1991	\$10.79
Dev Q5	2022Q3	\$2.00	1.1817	\$2.36

Calculate inflation-adjusted outstanding

Compute the outstanding $O_k(v)$ retrospectively at every quarter v it was open.

[< Prev](#)

Claim 10 of 20 (CLM-0001)

[Next >](#)

Ultimate = Total Payments Over Claim Lifetime (adjusted to 2025Q1) = \$55.12

Quarter	Cumulative Paid	Outstanding Liability
Dev Q1	\$0.00	\$55.12
Dev Q2	\$0.00	\$55.12
Dev Q3	\$0.00	\$55.12
Dev Q4	\$10.88	\$44.24
Dev Q5	\$13.26	\$41.86
Dev Q6	\$20.28	\$34.84
Dev Q7	\$22.59	\$32.53
Dev Q8	\$26.00	\$29.12
Dev Q9	\$55.12	\$0.00
Dev Q10	\$55.12	\$0.00

True outstanding = Ultimate – CumulativePaidToDate

From history to features

Everything known about claim k at valuation quarter v :

$$\mathcal{H}_{k,v} := \{A_k, N_k, (Z_{k,t})_{t \leq v}, (P_{k,t})_{t \leq v}, (C_{k,t})_{t \leq v}\}$$

Some covariates $Z_{k,t}$ are *static* (age at accident, injury severity), others are *time-varying* (legal representation can switch on mid-claim).

Problem: this history has a *different length* for every (k, v) . Tabular neural networks want a fixed-length vector.

The preprocessing pipeline is a deterministic map

$$X_{k,v} := \Phi(\mathcal{H}_{k,v}) \in \mathbb{R}^p.$$

Two kinds of time-varying covariates

When Φ compresses the history, the time-varying covariates split into two kinds:

1. *Only the latest value matters.* E.g. legal representation status, or the current case estimate: keep $Z_{k,v}$, discard the path. (Implicitly a *Markov-style* assumption: the current state is a sufficient summary.)
2. *The path matters.* E.g. the payments-per-quarter series: a claim that paid \$40 in one early lump differs from one that dribbled out \$10 four times. These get summarised as functions of the whole history — they are *path-dependent* features.

Deciding which kind each covariate is *is itself a modelling choice*: treating the case estimate as “latest value only” throws away how often the assessor has revised it.

Summarising the payment history

How does Φ compress a variable-length payment history into fixed columns? Summary statistics:

- development quarter $v - A_k$, notification delay $N_k - A_k$,
- count / sum / mean / std / min / max of past payments,
- the latest case estimate,
- current values of the time-varying covariates,
- the static covariates unchanged.

← Prev

Claim 10 of 20 (CLM-0001)

Next →

Choose development cutoff:

Dev Q2

Dev Q11

Cutoff: Dev Q6 2022Q3

Near-Static Covariates (latest up to cutoff)

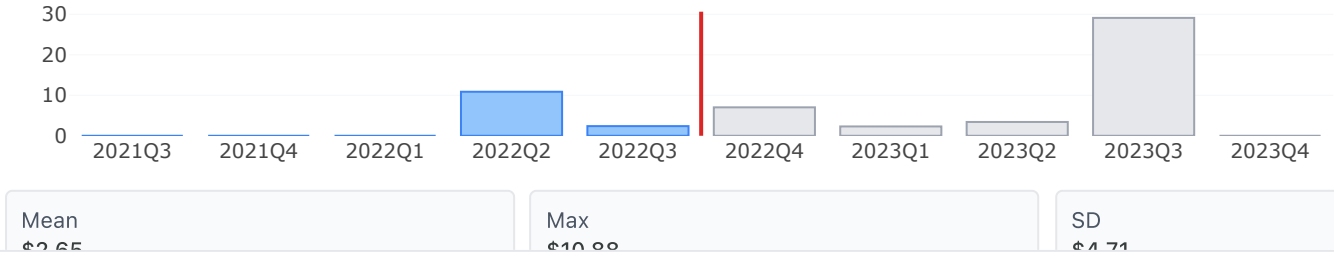
Dev Q	Calendar Q	Postcode	Legal Rep
Q2	2021Q3	1000	NA
Q3	2021Q4	1000	NA
Q4	2022Q1	1000	Yes
Q5	2022Q2	1000	Yes
Q6	2022Q3	1000	Yes
Q7	2022Q4	1000	Yes

Most recent Postcode
1000

Claimant has legal representation
Yes

Time-Series Covariates (summary up to cutoff)

Incremental payments



[< Prev](#)

Claim 10 of 20 (CLM-0001)

[Next >](#)

Choose development cutoff:

Dev
Q2

Cutoff: Dev Q6 2022Q3

Dev
Q11

Training Row (as-of Dev Q6)

Feature	Type	Rule	Value
postcode_latest	Near-static	Latest non-missing $\leq$ cutoff	1000
legal_rep_latest	Near-static	Latest non-missing $\leq$ cutoff	Yes
inc_paid_mean_q1..k	Time-series	Mean of increments $\leq$ cutoff	\$2.65
inc_paid_max_q1..k	Time-series	Max of increments $\leq$ cutoff	\$10.88
inc_paid_sd_q1..k	Time-series	Std dev of increments $\leq$ cutoff	\$4.71
cum_paid_last_qk	Time-series	Last observed $\leq$ cutoff	\$13.26
case_estimate	Time-series	Case estimate (latest $\leq$ cutoff)	\$44.40
outstanding	Target	Actual outstanding = ultimate – paid-to-date	\$41.86

Alternative: let an RNN read the history

The claim history is naturally a *sequence* — one vector per development quarter:

$$(P_{k,t}, C_{k,t}, Z_{k,t}) \quad \text{for } t = N_k, \dots, v.$$

So instead of hand-crafting summary statistics, we could:

- feed the quarter-by-quarter vectors into a recurrent layer (e.g. GRU/LSTM),
- take the final hidden state as a *learned* summary of the history,
- concatenate the static covariates, and predict $O_k(v)$ with a dense head.

Preparing features for a neural network

Finally, add standard NN preprocessing ($\log(1 + x)$ transforming monetary values), encoding categoricals, normalising continuous features.

[< Prev](#)

Claim 10 of 20 (CLM-0001)

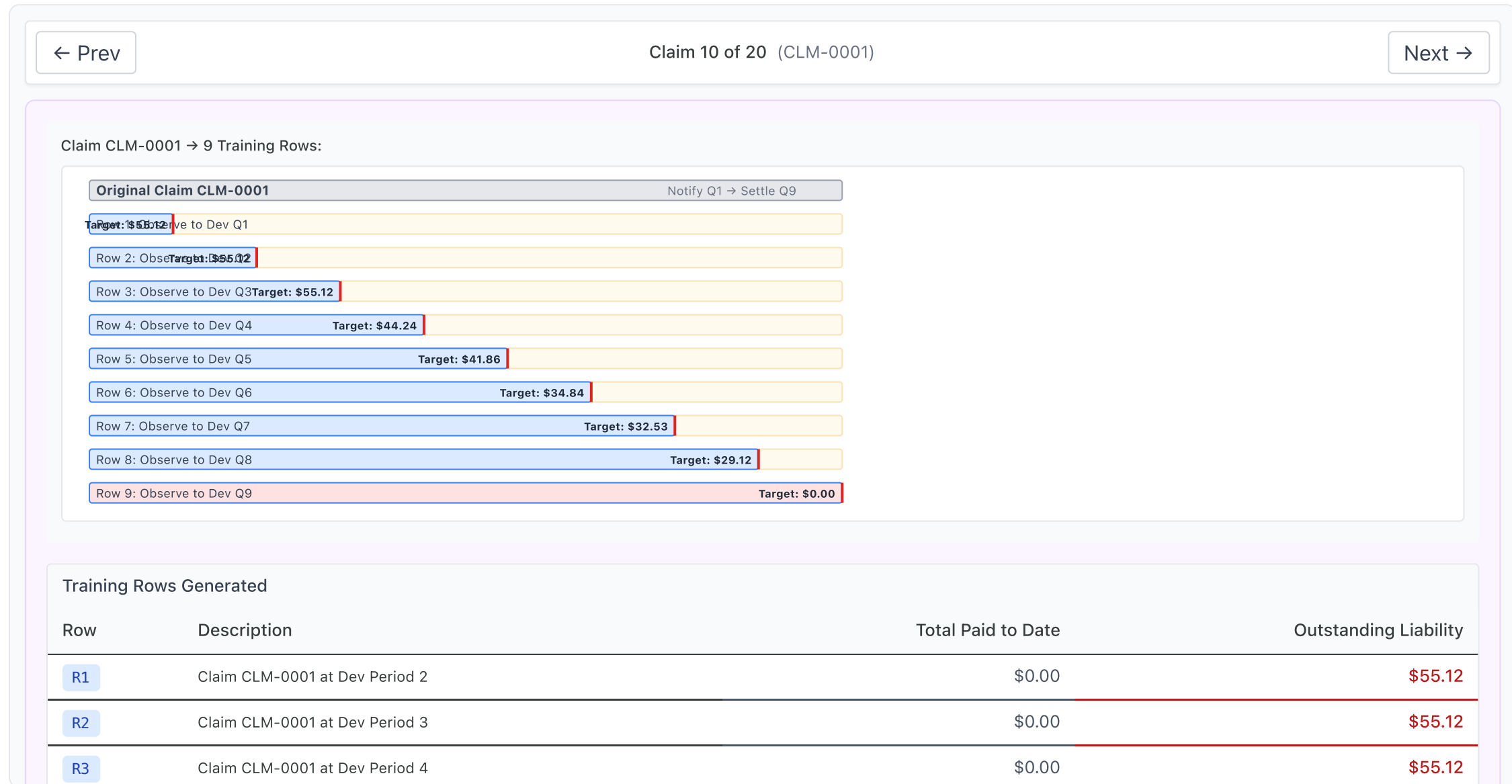
[Next >](#)

Training Row with Log-Transformed Dollar Features

Feature	Original Value	$\log(1 + x)$
postcode_latest	(categorical - will be one-hot encoded)	
legal_rep_latest	(categorical - will be one-hot encoded)	
inc_paid_mean_q1..k	\$2.65	1.2952
inc_paid_max_q1..k	\$10.88	2.4747
inc_paid_sd_q1..k	\$4.22	1.6516
cum_paid_last_qk	\$13.26	2.6574
case_estimate	\$37.45	3.6492
outstanding	\$41.86	

Creating claim snapshots

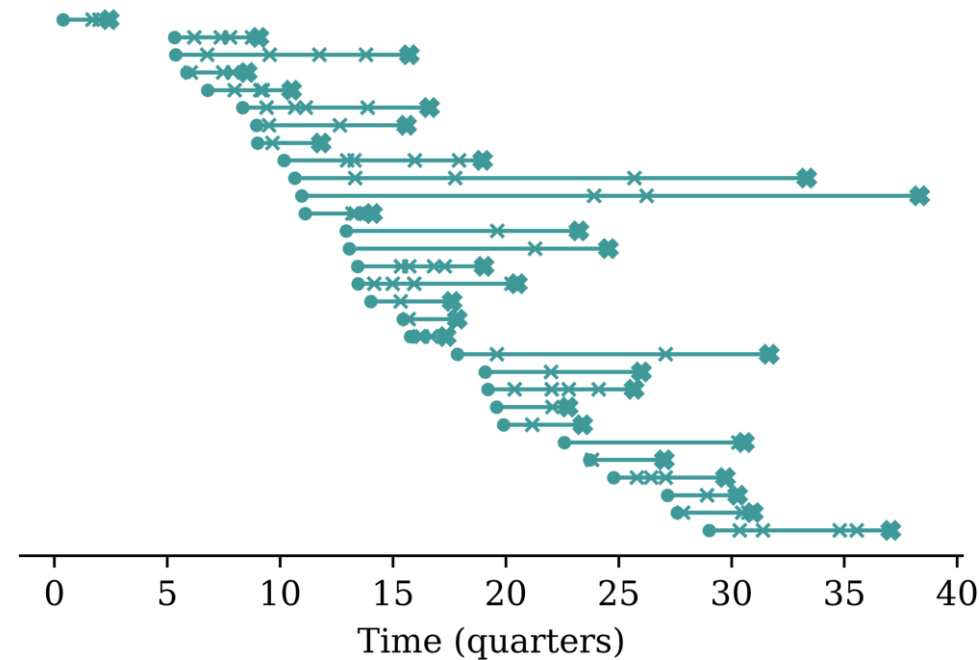
One snapshot row per development quarter, features $X_{k,v}$ with target $O_k(v)$.



Lecture Outline

- Introduction
- Insurance Claim Payments and Case Estimates
- Benchmark: Aggregate Reserving
- Individual Claim Modelling Choices
- Preprocessing One Claim
- **Splitting Claims by Finalisation Quarter**
- Splitting Claims by Notification Quarter
- Case Study: Synthetic Claims
- Fitting the Models
- Metrics
- Did the Model Beat the Assessors?
- Wrapping Up

A portfolio of claims over time

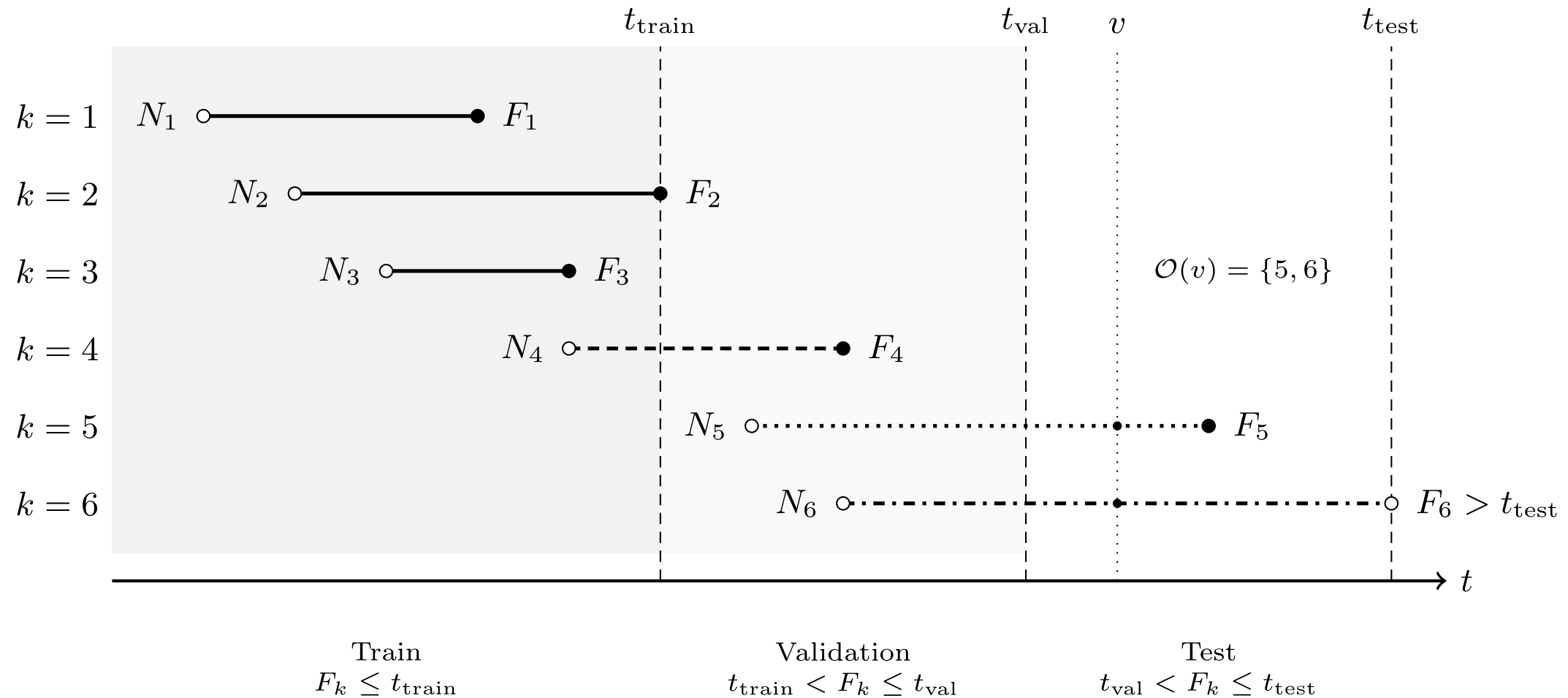


Now zoom out from one claim to the whole portfolio. Every claim needs to be allocated to train, validation, or test — but claims *overlap in time*, so the familiar shuffled random split deserves suspicion.

Remember, claim k generates $F_k - N_k$ snapshot rows, and consecutive snapshots of the same claim are *near-duplicates* (the history only grows by one quarter). So: *all snapshots from one claim go to the same set*.

The simplest approach

Order claims by finalisation quarter, then cut into *train*, *validation*, and *test* eras:

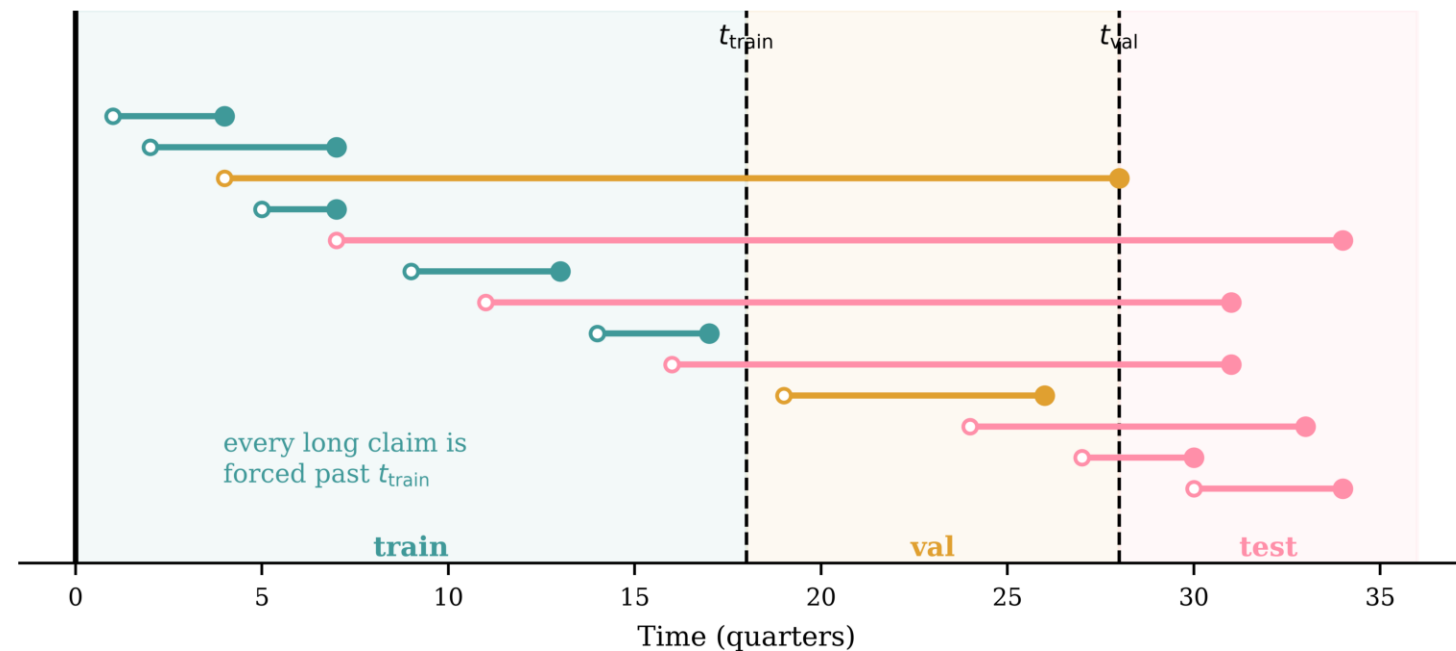


Those still open at t_{test} are the *unfinalised* set.

Old claims skew a naive split

Suppose we keep every claim *notified* after the portfolio's inception — the obvious thing to do for a *brand-new* insurer. A hard wall then sits at the start: no claim reaches back across it.

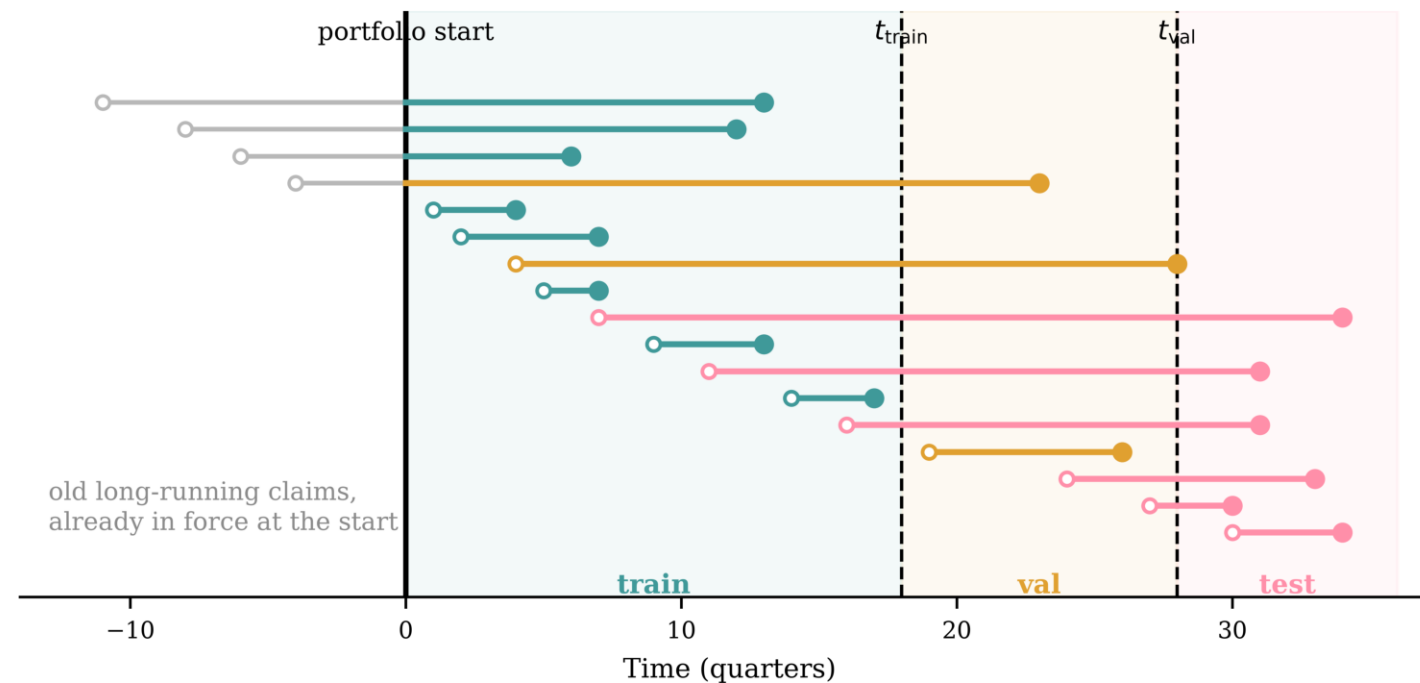
A claim can only finalise in the *train* era if it both *arrived* and *closed* early, which caps its duration at t_{train} . Every genuinely long-running claim is forced past the cut, into *validation* or *test*.



The *train* set never sees a long claim, yet the model is judged on them — a mismatch

Fix: arrive at a steady-state portfolio

A real reserving team never starts at inception; they inherit a portfolio already *in force*, full of claims opened years earlier. We mimic that: simulate a long *burn-in* of history first, then keep every claim *finalised* after a chosen start date (not every claim *notified* after it).

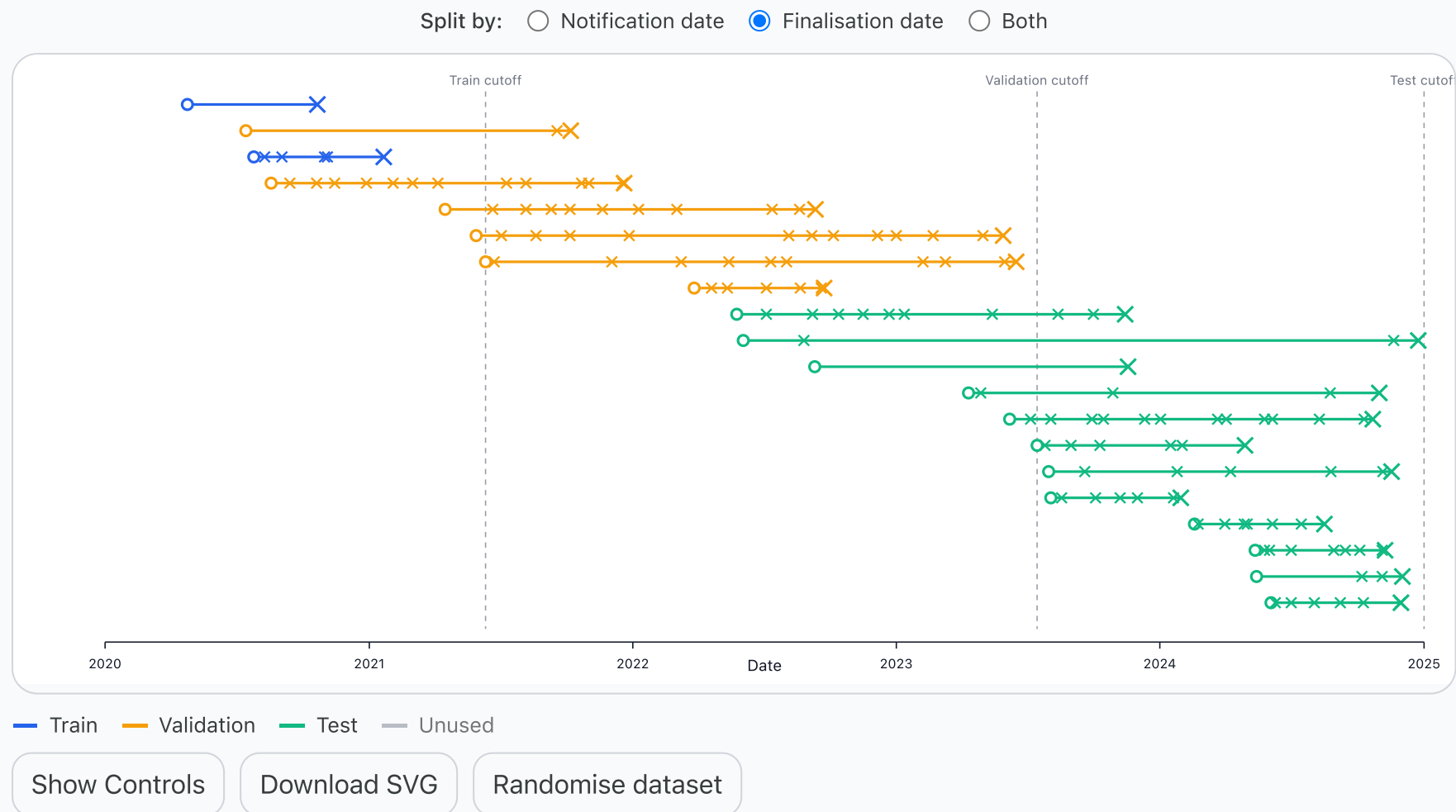


Old long-running claims now finalise inside the *train* era too, so every split carries the full spread of durations. We simulate many quarters and place the observation window deep in the steady state.

Lecture Outline

- Introduction
- Insurance Claim Payments and Case Estimates
- Benchmark: Aggregate Reserving
- Individual Claim Modelling Choices
- Preprocessing One Claim
- Splitting Claims by Finalisation Quarter
- **Splitting Claims by Notification Quarter**
- Case Study: Synthetic Claims
- Fitting the Models
- Metrics
- Did the Model Beat the Assessors?
- Wrapping Up

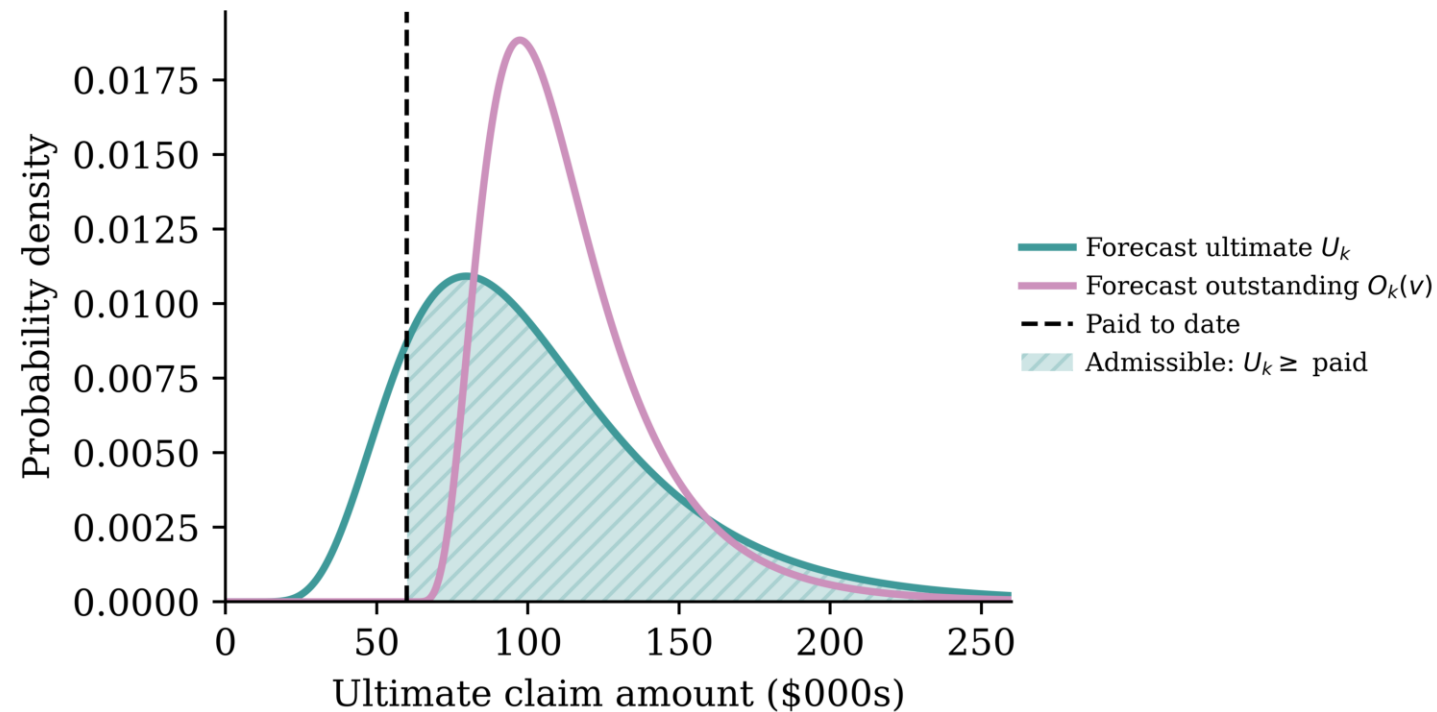
This is more complex



Never-finalising claims are censored

If the claim k is still open at the end of our dataset, then we have not observed the ultimate.

We only know a lower bound, $U_k \geq P_k(v)$, i.e. it is a *right-censored* observation (the survival-analysis kind).



The ultimate model must put *zero* mass left of the dashed line. The outstanding model gets this free: it lives on $[0, \infty)$ from the start.

Training directly on the censored ultimate

Keep the ultimate U_k as the target and borrow the *censored likelihood* from survival analysis.

- $\delta_k \in \{0, 1\}$: claim k is *finalised* ($\delta_k = 1$) or still *open* ($\delta_k = 0$).
- $P_k(v) = \sum_{t \leq v} P_{k,t}$: the *paid-to-date* for claim k at the training cut v .
- Distributional regression: $U_k \mid X_k \sim \mathcal{D}(\vartheta_k)$ with $\vartheta_k = \vartheta(X_k; \theta)$, giving density f_k , cdf F_k , and survival $S_k = 1 - F_k$.

What we actually observe:

- *Finalised* ($\delta_k = 1$): the exact ultimate $U_k = u_k$.
- *Open* ($\delta_k = 0$): only the bound $U_k \geq P_k(v)$ (right-censored at $P_k(v)$).

The censored log-likelihood

A finalised claim contributes its *density* at u_k ; an open claim contributes its *survival* at $P_k(v)$ (all we know is $U_k \geq P_k(v)$):

$$L(\theta) = \prod_{k=1}^n [f_k(u_k)]^{\delta_k} [S_k(P_k(v))]^{1-\delta_k}.$$

The loss is the negative log-likelihood:

$$-\ell(\theta) = -\sum_{k=1}^n \left[\delta_k \log f_k(u_k) + (1 - \delta_k) \log (1 - F_k(P_k(v))) \right].$$

Open claims still inform θ through the survival term $S_k(P_k(v))$: the constraint $U_k \geq P_k(v)$ is genuinely informative when the target is the ultimate.

But if we're predicting the outstanding...

We'll predict the *outstanding* $O_k(v)$ rather than the ultimate U_k directly.

- For a censored claim, the bound $U_k \geq P_k(v)$ becomes just $O_k(v) \geq 0$.
- Any sensible model of the outstanding only predicts *positive* values.
- So the censoring constraint is satisfied *by construction* — censored claims carry no usable information, and we can simply drop them.

Lecture Outline

- Introduction
- Insurance Claim Payments and Case Estimates
- Benchmark: Aggregate Reserving
- Individual Claim Modelling Choices
- Preprocessing One Claim
- Splitting Claims by Finalisation Quarter
- Splitting Claims by Notification Quarter
- **Case Study: Synthetic Claims**
- Fitting the Models
- Metrics
- Did the Model Beat the Assessors?
- Wrapping Up

Imports for the case study

The R side simulates the data and runs the classical benchmark:

```
1 library(SPLICE)
2 library(ChainLadder)
3 library(arrow)
```

The Python side does the machine learning:

```
1 import numpy as np
2 import pandas as pd
3 from sklearn.compose import ColumnTransformer
4 from sklearn.linear_model import Ridge
5 from sklearn.metrics import root_mean_squared_error
6 from sklearn.pipeline import make_pipeline
7 from sklearn.preprocessing import FunctionTransformer, MinMaxScaler, OneHotEncoder
8
9 from keras.models import Sequential
10 from keras.layers import Dense
11 from keras.callbacks import EarlyStopping
```

Generating synthetic claims with SPLICE

```

1  sim ← generate_data(
2    n_claims_per_period = 100,    # ~10,000 claims over the full horizon
3    n_periods           = 100,    # long burn-in; we observe the last ~40 quarters
4    complexity          = 2,      # development depends on claim size; 2% p.a. inflation
5    random_seed         = 20260712,
6    covariates_obj      = SynthETIC::test_covariates_obj
7  )
8
9  # SPLICE names the case-estimate column `OCL`; rename it to `estimate`.
10 names(sim$incurred_dataset)[names(sim$incurred_dataset) == "OCL"] ← "estimate"
11
12 claims ← cbind(sim$claim_dataset, sim$covariates_data$data)
13 write_parquet(claims, "data/splice/claims.parquet")
14 write_parquet(sim$payment_dataset, "data/splice/payments.parquet")
15 write_parquet(sim$incurred_dataset, "data/splice/estimates.parquet")

```

Three tables: one row per *claim*, per *payment*, and per *case estimate transaction*.

```

1  claims = pd.read_parquet("data/splice/claims.parquet")
2  payments = pd.read_parquet("data/splice/payments.parquet")
3  estimates = pd.read_parquet("data/splice/estimates.parquet")

```

All claims

1 claims

	claim_no	occurrence_period	occurrence_time	claim_size	notidel	setldel	no_payment	Legal Representati
0	1	1.0	0.797054	2.398522e+04	1.756252	6.321124	4.0	Y
1	2	1.0	0.045334	6.775792e+04	0.336995	9.308473	5.0	Y
2	3	1.0	0.532276	2.906999e+06	1.504237	22.077790	9.0	Y
...	...	...	...	...	...	...	...	...
9971	9972	100.0	99.703348	9.786867e+04	2.284232	5.694223	4.0	Y
9972	9973	100.0	99.186038	6.959615e+05	1.044376	12.717365	7.0	Y
9973	9974	100.0	99.190642	2.340260e+05	0.048471	21.621160	4.0	Y

9974 rows × 10 columns

All payments

1 payments

	claim_no	pmt_no	occurrence_period	occurrence_time	claim_size	notidel	setldel	payment_time
0	1	1.0	1.0	0.797054	23985.215278	1.756252	6.321124	3.433536
1	1	2.0	1.0	0.797054	23985.215278	1.756252	6.321124	7.205018
2	1	3.0	1.0	0.797054	23985.215278	1.756252	6.321124	7.770359
...	...	...	...	...	...	...	...	...
49608	9974	2.0	100.0	99.190642	234025.981387	0.048471	21.621160	112.338750
49609	9974	3.0	100.0	99.190642	234025.981387	0.048471	21.621160	118.956128
49610	9974	4.0	100.0	99.190642	234025.981387	0.048471	21.621160	120.860272

49611 rows × 12 columns

All estimates

1 estimates

	claim_no	claim_size	txn_time	txn_delay	txn_type	incurred	estimate	cumpaid	multipli
0	1	23985.215278	2.553306	0.000000	Ma	28490.680585	28490.680585	0.000000	1.000000
1	1	23985.215278	3.433536	0.880230	PMi	29751.957275	26249.028917	3502.928359	1.039729
2	1	23985.215278	5.770253	3.216947	Mi	30564.919549	27061.991190	3502.928359	1.017578
...	...	...	...	...	...	...	...	...	...
84836	9974	234025.981387	118.956128	19.717015	PMi	414317.471329	20937.327762	393380.143567	0.923684
84837	9974	234025.981387	119.646620	20.407507	Mi	415272.233409	21892.089843	393380.143567	0.977916
84838	9974	234025.981387	120.860272	21.621160	PMi	417673.416862	0.000000	417673.416862	0.995394

84839 rows × 9 columns

One synthetic claim's payments

One claim's payments (note the paired nominal/deflated columns):

```
1 payments[payments["claim_no"] == 1][ ... ]
```

	pmt_no	payment_time	payment_period	payment_size	payment_inflated
0	1.0	3.433536	4.0	3443.887887	3502.928359
1	2.0	7.205018	8.0	3634.989595	3766.988290
2	3.0	7.770359	8.0	15120.618241	15713.616650
3	4.0	8.874430	9.0	1785.719555	1865.922752

One synthetic claim's case estimates

Its case estimate transactions:

```
1 estimates[estimates["claim_no"] = 1][ ... ]
```

	txn_time	txn_type	cumpaid	estimate
0	2.553306	Ma	0.000000	28490.680585
1	3.433536	PMi	3502.928359	26249.028917
2	5.770253	Mi	3502.928359	27061.991190
...	...	...	...	...
6	7.770359	P	22983.533298	2119.659221
7	8.193496	Mi	22983.533298	1941.640369
8	8.874430	PMi	24849.456051	0.000000

9 rows × 4 columns

From transactions to snapshot rows

Bin each claim's key dates into quarters, and sort the claims into groups:

```

1  # We "arrive" at the portfolio at T_START and value it at T_END: keep claims
2  # that finalise inside this steady-state window, not those that merely start in it.
3  T_START, T_END = 60, 100
4
5  claims["notif_qtr"] = np.ceil(claims["occurrence_time"] + claims["notidel"]).astype(int)
6  claims["final_qtr"] = np.ceil(claims["occurrence_time"] + claims["notidel"] + claims["set
7
8  notif, final = claims["notif_qtr"], claims["final_qtr"]
9  ibnr = notif > T_END
10 unfinalised = (notif ≤ T_END) & (final > T_END)
11 flash = (final ≤ T_END) & (final == notif)
12 observed = claims[(final > T_START) & (final ≤ T_END) & (final > notif)]

```

9,974 claims: 3,935 finalised & usable, 971 unfinalised, 214 IBNR, 162 settled within a quarter

Building the snapshot grid

One row per claim per open quarter, with the quarterly payments $P_{k,t}$ merged in (missing quarters become zero-payment rows).

Before — one row per *claim*:

	claim_no	notif_qtr	final
2823	2824	31	61
2919	2920	31	65
2987	2988	32	63

After — one row per claim per open *quarter*:

	claim_no	quarter	Incremental
0	2824	31	0.00
1	2824	32	0.00
2	2824	33	0.00
3	2824	34	0.00
4	2824	35	24071.86
5	2824	36	0.00

History summaries, case estimates, and the target

Onto that grid we add the path-dependent features (expanding summaries of the payment history), the current-state features (latest case estimate, forward-filled between revisions).

Before — one claim's rows:

	quarter	Incremental
0	31	0.00
1	32	0.00
2	33	0.00
3	34	0.00
4	35	24071.86

After — summaries, case estimate, and the label $O_k(v)$:

	quarter	sum_Incremental	Estimate	Outstanding
0	31	0.00	106362.62	1573886.40
1	32	0.00	106362.62	1573886.40
2	33	0.00	106362.62	1573886.40
3	34	0.00	106362.62	1573886.40
4	35	24071.86	80251.97	1549814.53

The dataset after preprocessing

```
1 grid[["claim_no", "quarter", "development_period", "Incremental",
2      "sum_Incremental", "Estimate", "Outstanding"]].head(8)
```

	claim_no	quarter	development_period	Incremental	sum_Incremental	
0	2824	31	3.0	0.0	0.0000000	106
1	2824	32	4.0	0.0	0.0000000	106
2	2824	33	5.0	0.0	0.0000000	106
...	...	...	...	...	...	...
5	2824	36	8.0	0.0	24071.863793	802
6	2824	37	9.0	0.0	24071.863793	802
7	2824	38	10.0	0.0	24071.863793	802

8 rows × 7 columns

Splitting by finalisation quarter

```
1 T_TRAIN, T_VAL = 88, 94 # grid already starts at T_START, so train is (60, 88]
2 X_train = grid[grid["final_qtr"] ≤ T_TRAIN]
3 X_val = grid[(grid["final_qtr"] > T_TRAIN) & (grid["final_qtr"] ≤ T_VAL)]
4 X_test = grid[grid["final_qtr"] > T_VAL]
```

train	28,992 rows	2,790 claims	(finalisation quarters 61..88)
val	6,273 rows	598 claims	(finalisation quarters 89..94)
test	5,593 rows	547 claims	(finalisation quarters 95..100)

Note rows $\gg$ claims: each claim contributes one row per quarter it was open — and every one of a claim's rows lands in the same set.

Lecture Outline

- Introduction
- Insurance Claim Payments and Case Estimates
- Benchmark: Aggregate Reserving
- Individual Claim Modelling Choices
- Preprocessing One Claim
- Splitting Claims by Finalisation Quarter
- Splitting Claims by Notification Quarter
- Case Study: Synthetic Claims
- **Fitting the Models**
- Metrics
- Did the Model Beat the Assessors?
- Wrapping Up

Benchmark: chain ladder

```

1 payments <- read_parquet("data/splice/payments.parquet")
2 # The triangle uses the observed window (accident quarters 61-100), shifted to
3 # 1-40 so it reads as ten fresh accident years valued at quarter 100.
4 window <- subset(payments, occurrence_period > 60 & payment_period ≤ 100)
5 window$acc_year <- ceiling((window$occurrence_period - 60) / 4)
6 window$dev_year <- ceiling((window$payment_period - 60) / 4) -
7   window$acc_year + 1
8
9 incr <- aggregate(payment_size ~ acc_year + dev_year, window, sum)
10 tri <- incr2cum(as.triangle(incr, origin = "acc_year",
11                             dev = "dev_year", value = "payment_size"))
12 round(tri / 1e6, 2) # cumulative payments, $m

```

acc_year	dev_year								
	1	2	3	4	5	6	7	8	9
1	0.30	3.09	9.09	18.92	33.61	44.07	56.21	64.68	69.82
2	0.48	3.92	12.84	28.16	41.98	59.02	68.39	74.46	76.80
3	0.32	3.65	10.06	18.23	31.71	47.09	48.47	50.93	NA
4	0.38	4.43	12.12	26.74	40.39	53.48	64.34	NA	NA
5	0.31	4.36	10.41	24.81	41.64	55.46	NA	NA	NA
6	0.31	3.41	11.87	21.74	35.67	NA	NA	NA	NA
7	0.27	3.40	9.45	17.67	NA	NA	NA	NA	NA
8	0.33	3.45	10.43	NA	NA	NA	NA	NA	NA
9	0.35	3.17	NA	NA	NA	NA	NA	NA	NA
10	0.36	NA	NA	NA	NA	NA	NA	NA	NA

The chain ladder reserve

```
1 mack ← MackChainLadder(tri, est.sigma = "Mack")
2 reserve_cl ← summary(mack)$Totals["IBNR:", ]
3 true_outstanding ← sum(subset(payments, payment_period > 100)$payment_size)
4
5 cat(sprintf("Chain ladder reserve:   $%.1fm\n", reserve_cl / 1e6),
6     sprintf("True total outstanding: $%.1fm\n", true_outstanding / 1e6))
```

Chain ladder reserve: \$275.2m
True total outstanding: \$290.8m

As this is synthetic data, we know the future and can compare against the true value.

A baseline model: GLM

We model $\log O_k(v)$ rather than $O_k(v)$ because the outstanding is *heavy-tailed*.

Squared-error loss on $\log O_k(v)$ is then a *lognormal* assumption for the outstanding, so every RMSE below is on the *log scale*, not in dollars.

```

1 dollar_cols = ["Incremental", "sum_Incremental", "mean_Incremental",
2               "std_Incremental", "min_Incremental", "max_Incremental"]
3 numeric_cols = ["development_period", "notidel", "count_Incremental"]
4 cat_cols = ["Legal Representation", "Injury Severity", "Age of Claimant"]
5
6 def make_preprocessor(with_estimate):
7     dollars = dollar_cols + (["Estimate"] if with_estimate else [])
8     return ColumnTransformer([
9         ("dollars", make_pipeline(FunctionTransformer(np.log1p),
10                                  MinMaxScaler()), dollars),
11         ("numeric", MinMaxScaler(), numeric_cols),
12         ("categorical", OneHotEncoder(drop="first", sparse_output=False),
13          cat_cols),
14     ])

```

Two versions: *without* and *with* the case estimate as a feature.

Setup X & y variables

```
1 y_train = np.log(X_train["Outstanding"])
2 y_val = np.log(X_val["Outstanding"])
3 y_test = np.log(X_test["Outstanding"])
4
5 ct_plain = make_preprocessor(with_estimate=False)
6 train_plain = ct_plain.fit_transform(X_train)
7 val_plain = ct_plain.transform(X_val)
8 test_plain = ct_plain.transform(X_test)
9
10 ct_est = make_preprocessor(with_estimate=True)
11 train_est = ct_est.fit_transform(X_train)
12 val_est = ct_est.transform(X_val)
13 test_est = ct_est.transform(X_test)
```

GLM

```

1 glm_plain = Ridge(alpha=1e-4).fit(train_plain, y_train)
2 glm_est = Ridge(alpha=1e-4).fit(train_est, y_train)
3
4 def print_rmse(label, pred):
5     print(f"{label:34s} test RMSE (log scale) = "
6           f"{root_mean_squared_error(y_test, pred):.3f}")
7
8 print_rmse("GLM without case estimates", glm_plain.predict(test_plain))

```

GLM without case estimates test RMSE (log scale) = 1.277

```

1 print_rmse("GLM with case estimates", glm_est.predict(test_est))

```

GLM with case estimates test RMSE (log scale) = 1.029

```

1 print_rmse("Case estimates alone", np.log(X_test["Estimate"]))

```

Case estimates alone test RMSE (log scale) = 1.418

Fitting a neural network

```
1 # Reuse the "with case estimate" matrices prepared for the GLM above.
2 nn = Sequential([
3     Dense(30, activation="leaky_relu"),
4     Dense(30, activation="leaky_relu"),
5     Dense(1)])
6 nn.compile(optimizer="adam", loss="mse")
7 es = EarlyStopping(patience=10, restore_best_weights=True)
8 hist = nn.fit(train_est, y_train, epochs=100,
9               validation_data=(val_est, y_val), callbacks=[es], verbose=0)
```


Lecture Outline

- Introduction
- Insurance Claim Payments and Case Estimates
- Benchmark: Aggregate Reserving
- Individual Claim Modelling Choices
- Preprocessing One Claim
- Splitting Claims by Finalisation Quarter
- Splitting Claims by Notification Quarter
- Case Study: Synthetic Claims
- Fitting the Models
- **Metrics**
- Did the Model Beat the Assessors?
- Wrapping Up

Evaluating the models

All three models on the test set, against the benchmark-to-beat:

```
1 pred_plain = glm_plain.predict(test_plain)
2 pred_est = glm_est.predict(test_est)
3 pred_nn = nn.predict(test_est, verbose=0).flatten()
4 pred_case = np.log(X_test["Estimate"])
5
6 print_rmse("GLM without case estimates", pred_plain)
```

GLM without case estimates test RMSE (log scale) = 1.277

```
1 print_rmse("GLM with case estimates", pred_est)
```

GLM with case estimates test RMSE (log scale) = 1.029

```
1 print_rmse("Neural network with case estimates", pred_nn)
```

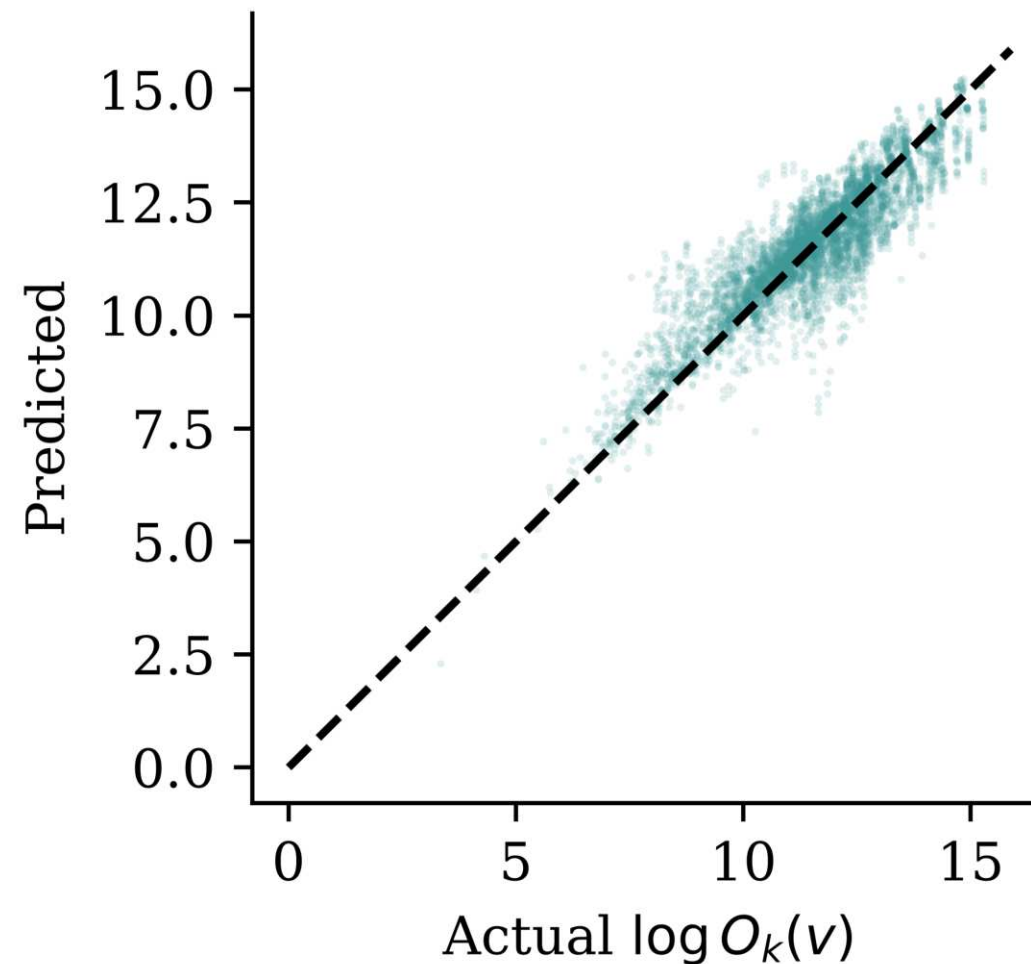
Neural network with case estimates test RMSE (log scale) = 0.721

```
1 print_rmse("Case estimates alone", pred_case)
```

Case estimates alone test RMSE (log scale) = 1.418

Per-snapshot accuracy is one lens — but a reserving model's real job is the portfolio total.

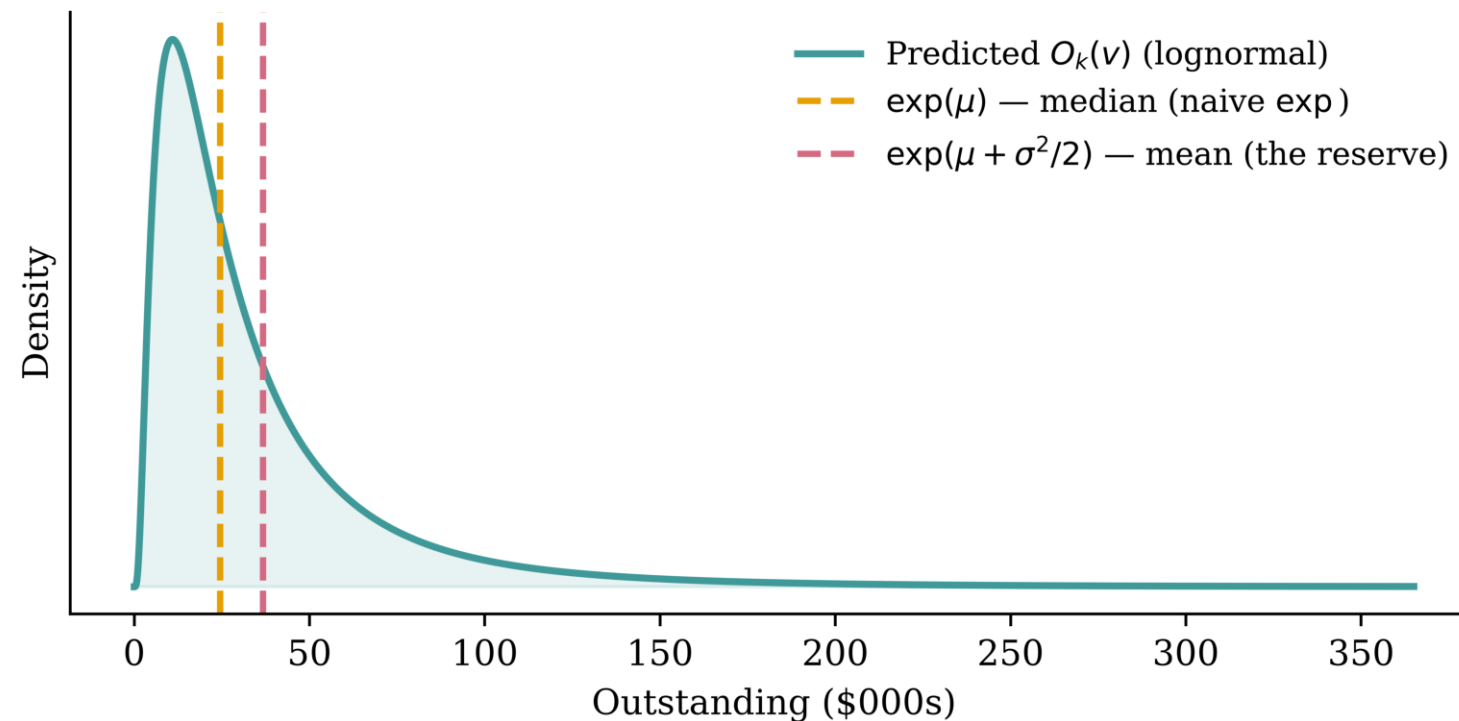
Predicted vs actual outstanding



Each point is one test-set snapshot $(X_{k,v}, O_k(v))$, plotted on the model's log scale; the dashed diagonal is perfect prediction.

Exponentiating log predictions under-reserves

Squared-error loss on $\log O_k(v)$ is exactly the constant-variance normal assumption from distributional regression, $\log O_k(v) \sim \mathcal{N}(\mu_k, \sigma^2)$, so on the dollar scale $O_k(v)$ is *lognormal*. The naive $\exp(\mu_k)$ is its *median*; a reserve needs the *mean*, $\exp(\mu_k + \frac{1}{2}\sigma^2)$.



Dropping the $\frac{1}{2}\sigma^2$ term predicts the median and *systematically under-reserves*; we add it back, estimating σ by the log-scale RMSE.

From per-claim predictions to the reserve

Build one feature row per still-open claim (X_{open}), predict $\hat{O}_k(v)$, and sum to the portfolio reserve $\hat{T}(v)$.

```

1 open_plain = ct_plain.transform(X_open)      # the same two recipes as for training
2 open_est = ct_est.transform(X_open)
3
4 def to_dollars(pred_log, sigma2):            # lognormal mean, exp(mu + sigma^2/2)
5     return np.exp(pred_log + sigma2 / 2)
6
7 # sigma^2 per model: its squared validation RMSE on the log scale.
8 sigma2_plain = np.mean((y_val - glm_plain.predict(val_plain)) ** 2)
9 sigma2_est = np.mean((y_val - glm_est.predict(val_est)) ** 2)
10 sigma2_nn = np.mean((y_val - nn.predict(val_est, verbose=0).flatten()) ** 2)
11
12 reserves = {
13     "Chain ladder": float(r.reserve_cl),
14     "GLM": to_dollars(glm_plain.predict(open_plain), sigma2_plain).sum(),
15     "GLM + case est.": to_dollars(glm_est.predict(open_est), sigma2_est).sum(),
16     "Neural net + case est.": float(to_dollars(
17         nn.predict(open_est, verbose=0).flatten(), sigma2_nn).sum()),
18 }
```

The portfolio reserves compared



The individual models can only reserve for claims they can *see*, so their fair target is the lower value (left line); the chain ladder triangle implicitly projects the IBNR claims too, so its target is the larger value (right line).

Lecture Outline

- Introduction
- Insurance Claim Payments and Case Estimates
- Benchmark: Aggregate Reserving
- Individual Claim Modelling Choices
- Preprocessing One Claim
- Splitting Claims by Finalisation Quarter
- Splitting Claims by Notification Quarter
- Case Study: Synthetic Claims
- Fitting the Models
- Metrics
- **Did the Model Beat the Assessors?**
- Wrapping Up

Better%: how often do we win?

RMSE scores each model in isolation. A claims manager asks something *head-to-head*: for each open claim, is *our* reserve closer to the eventual payout than the *assessor's* case estimate?

Better% is the share of test claims we win. “Closer” is judged *proportionally*, on the *log scale* — being \$10k out on a \$2m claim is fine, on a \$12k claim a disaster. Above 50% and we win more often than not.

```

1 actual = X_test["Outstanding"]
2 # How far each forecast is from the truth, proportionally (a log-scale gap).
3 case_gap = np.abs(np.log(X_test["Estimate"] / actual)) # the assessor's gap
4
5 def beats_assessor(pred_log, sigma2):
6     model_gap = np.abs(np.log(to_dollars(pred_log, sigma2) / actual))
7     return model_gap < case_gap
8
9 wins_plain = beats_assessor(pred_plain, sigma2_plain)
10 wins_est = beats_assessor(pred_est, sigma2_est)
11 wins_nn = beats_assessor(pred_nn, sigma2_nn)
12
13 print(f"GLM without case estimates          Better% = {wins_plain.mean():5.1%}")

```

GLM without case estimates Better% = 34.3%

```

1 print(f"GLM with case estimates          Better% = {wins_est.mean():5.1%}")

```


Weighted Better%

The portfolio reserve is dominated by the *big* claims, so weight each win by the claim's ultimate — winning a \$2m claim counts for more than winning a \$5k one:

$$\text{Weighted Better\%} = \sum_k w_k \mathbf{1}\{\text{model wins claim } k\}, \quad w_k = \frac{U_k}{\sum_j U_j}.$$

```
1 weight = X_test["claim_no"].map(ultimate)
2 weight = weight / weight.sum()
3
4 print(f"GLM without case estimates      Weighted Better% = "
5       f"{(weight * wins_plain).sum():5.1%}")
```

GLM without case estimates Weighted Better% = 51.0%

```
1 print(f"GLM with case estimates          Weighted Better% = "
2       f"{(weight * wins_est).sum():5.1%}")
```

GLM with case estimates Weighted Better% = 51.6%

```
1 print(f"Neural network with case estimates Weighted Better% = "
2       f"{(weight * wins_nn).sum():5.1%}")
```

Neural network with case estimates Weighted Better% = 58.8%

Weighting by size lifts every model: they may trail the assessor on a plain *count* of claims,

Lecture Outline

- Introduction
- Insurance Claim Payments and Case Estimates
- Benchmark: Aggregate Reserving
- Individual Claim Modelling Choices
- Preprocessing One Claim
- Splitting Claims by Finalisation Quarter
- Splitting Claims by Notification Quarter
- Case Study: Synthetic Claims
- Fitting the Models
- Metrics
- Did the Model Beat the Assessors?
- **Wrapping Up**

Individual vs aggregate reserving

- *Aggregate* (chain ladder on triangles): simple, stable, industry standard; but throws away claim-level covariates and gives no per-claim view.
- *Individual*: predicts $O_k(v)$ per open claim; the portfolio reserve is just the sum $T(v)$; supports triage and segmentation; per-claim errors partially cancel in the aggregate.
- Individual level reserving is a data engineering headache, and the total still misses IBNR, so an individual model alone is *not yet* a full portfolio reserve.

What we didn't cover

- *IBNR*: our model only reserves claims it can see; incurred-but-not-reported claims need a separate (usually aggregate) adjustment.
- *Uncertainty*: a reserve needs a risk margin, not just a point estimate; distributional forecasts and ensembling give prediction intervals.
- *When does granular beat aggregate?*

Python package versions

```
1 from watermark import watermark
2 print(watermark(python=True, packages="keras,matplotlib,numpy,pandas,sklearn,torch"))
```

Python implementation: CPython

Python version : 3.14.5

IPython version : 9.15.0

keras : 3.15.0

matplotlib: 3.11.0

numpy : 2.5.1

pandas : 3.0.3

sklearn : 1.9.0

torch : 2.12.1

R package versions

```
1 cat(sessionInfo()$R.version$version.string, "\n\n")
```

R version 4.6.0 (2026-04-24)

```
1 for (p in c("SPICE", "Synthetic", "ChainLadder", "arrow", "reticulate", "knitr"))  
2   cat(sprintf("%-12s %s\n", p, as.character(packageVersion(p))))
```

SPICE	1.1.2
Synthetic	1.1.1
ChainLadder	0.2.21
arrow	24.0.0
reticulate	1.46.0
knitr	1.51

References

- Avanzi, B., Taylor, G., & Wang, M. (2023). SPLICE: A synthetic paid loss and incurred cost experience simulator. *Annals of Actuarial Science*, 17(1), 7–35. <https://doi.org/10.1017/S1748499522000057>
- Avanzi, B., Taylor, G., Wang, M., & Wong, B. (2021). SynthETIC: An individual insurance claim simulator with feature control. *Insurance: Mathematics and Economics*, 100, 296–308. <https://doi.org/10.1016/j.insmatheco.2021.06.004>